

NICKOLAS SIMONS

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EDUCATION

University of Illinois at Urbana-Champaign

Bachelor of Science in Computer Science

Minor in Game Studies and Design

Urbana, IL

Aug. 2022 – May 2025

GPA: 3.96

EXPERIENCE

The stu/dio at Illinois

Programming Lead

Urbana, IL

Aug. 2024 – Dec. 2025

- Led a team of 12 engineers across 8 sponsored projects, conducting code reviews and providing technical feedback to ensure code quality
- Maintained Perforce infrastructure and integrated CI pipelines using Horde and Jenkins for automated builds and reviews
- Established and enforced coding standards and development workflows to improve code maintainability across projects

Gameplay Programmer

April 2024 – Dec. 2025

- Designed and implemented gameplay features and tools for VR titles using C++ and Blueprints, with an emphasis on accessibility, performance, and usability in educational environments
- Debugged and resolved critical gameplay issues affecting interaction flow, core mechanics, and level progression
- Optimized gameplay code and materials to meet VR performance targets using profiling tools like Unreal Insights

PROJECTS

Master Dancer VR | UE5, C++, Blueprint

May 2024 – Nov 2025

- Built three quantized movement-based rhythm games end-to-end, developing core gameplay architecture using C++ and Blueprint
- Created data-driven level authoring tools to streamline designer workflows and support rapid iteration
- Implemented a system for anchoring and rendering subtitles in 3D space for VR applications, with localization support
- Developed reusable gameplay systems with a focus on performance, usability, and extensibility

VRchaeology | UE5, C++, Blueprint

May 2025 – Oct 2025

- Designed and implemented a modular quest system and in-editor tooling in C++ to guide user actions through reusable scripting logic and contextual prompts.
- Developed an interactive, tool-based VR excavation system in C++, simulating surface digging and stratigraphic layer analysis
- Implemented shared sublevel instances in UE5 to maintain continuity of actors and state across multiple uses of the same sublevel, ensuring consistent gameplay and scene behavior

Multiplayer Action Game | UE5, C++, Blueprint, ASIO, C#, GraphQL, PostgreSQL

Nov. 2025 – Present

- Built a multiplayer combat framework using C++ and Epic's Gameplay Ability System, enabling responsive and synchronized player interactions
- Developed a custom matchmaking utility microservice with C#, GraphQL, and PostgreSQL, integrating with Epic Online Services to improve matchmaking reliability and flexibility
- Created a system for replicated dynamic environment partitions using Epic's Procedural Content Generation framework, enhancing dynamic multiplayer gameplay.

Card/Equipment Battler | UE5, C++, Blueprint

Jan 2024 – Jun 2024

- Implemented lightweight effect-based card and item combat system with card resource restriction features
- Created tooling to fetch and generate card assets and enemies from design spreadsheets
- Developed procedural link-based dungeon generation system

Othello | Haskell, Python, OpenCV

May 2022

- Developed a variable-size Othello game with AI opponent using a minimax algorithm and pruned game tree in Haskell
- Built a computer vision application to detect Othello board state from images using the Hough transform

TECHNICAL SKILLS

Languages: C++, C, SQL(PostgreSQL), GraphQL, Python, Haskell, C#, Typescript, OpenGL

Developer Tools: Unreal Engine, Perforce, Git, Visual Studio, Unity, Docker, Horde, Jenkins

Libraries: ASIO (C++), Hot Chocolate(GraphQL), OpenCV, SlateIM, Numpy, Matplotlib